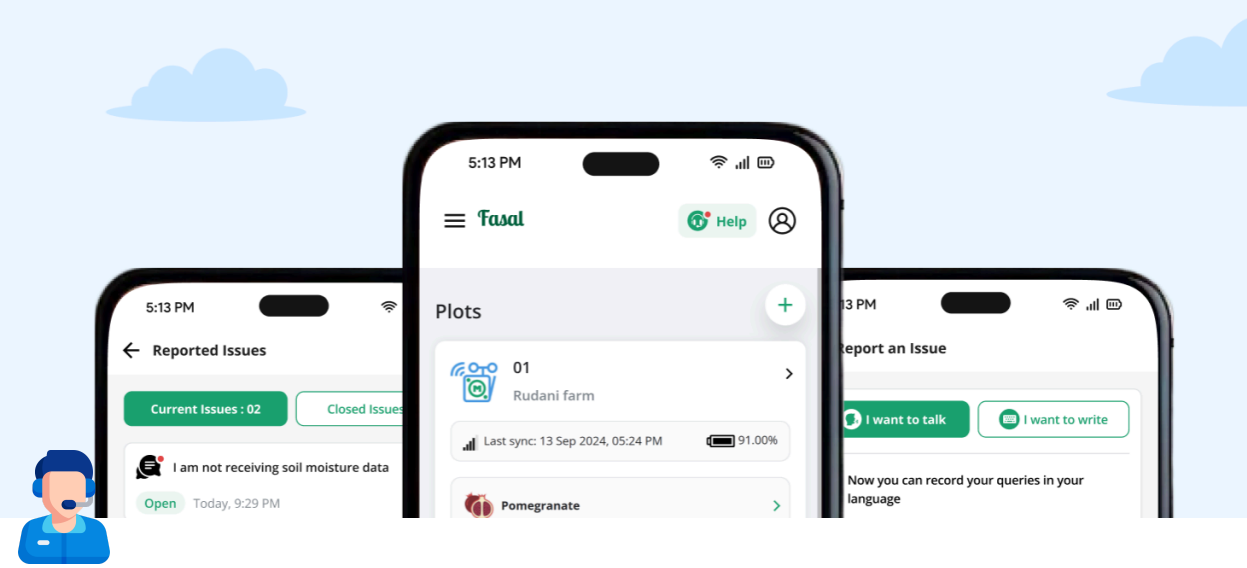


Fasal Support System: Reducing Support Tickets by 80%

Industry	B2B SaaS
Designation	Senior Product Designer
Focus areas	Cost Reduction, Operational Efficiency, Service Design, Support Automation
Tools Used	Figma, Google Docs, Google Sheets, Useberry

About Fasal app

Fasal is an innovative agricultural technology solution that leverages IoT devices and AI to revolutionize farming practices. Available on both iOS and Android, here's a brief overview of how the Fasal app works:

- IoT Sensors: Remote sensors placed on farms collect real-time data on crop, soil, and weather conditions.
- AI-Powered Analysis: The collected data is analyzed using artificial intelligence and machine learning algorithms.
- Personalized Insights: The app generates farm-specific, crop-specific, and crop-stage-specific insights.
- Local Language Support: These insights are delivered to farmers in their local languages through the Fasal app.
- Actionable Recommendations: Farmers receive timely advice on irrigation, pest control, fertilizer application, and other crucial aspects of crop management.

By providing these data-driven insights, Fasal empowers farmers to make informed decisions, optimize their crop growth conditions, and ultimately improve their agricultural productivity.

Project Summary: We revamped the Help Section in the Fasal App to improve farmers' ability to report issues. We simplified the process, added an audio recording feature for easier communication, and enhanced overall usability.

Key Improvements:

- Reduced steps from 6 to 4 in the help process
- Added audio recording feature to overcome language and literacy barriers
- Redesigned interface for better discoverability and user experience

Impact:

- **24.9%** increase in tickets submitted through mobile devices
- **107.6%** increase in tickets raised, showing improved user engagement due to better mobile accessibility and the new audio ticketing feature
- **Over a third of users (34.28%)** now use the audio ticketing feature, confirming that language barriers was a real issue.

My Role

As Senior Product Designer, I led the revamp of the Fasal App's Help Section. My key responsibilities included:

- Conducting comprehensive user research, including stakeholder interviews and field visits
- Analyzing data findings and prioritizing user pain points
- Defining problem statements and "How Might We" questions
- Developing multiple design solutions and creating interactive prototypes
- Executing user testing with farmers and implementing feedback
- Collaborating with cross-functional teams for successful implementation and impact measurement

Stakeholder Interviews, Data Findings & Field Visit Insights

We interviewed internal stakeholders, including the sales and customer success teams, to gain a deeper understanding of the problems farmers were facing. This allowed us to make an informed decision.

Stakeholders key findings

- Both teams emphasized the need for a more user-friendly help system that caters to farmers' preferences and limitations
- The importance of voice-based communication was consistently highlighted across both teams
- Language and literacy barriers were identified as significant obstacles in the current text-based help system

- Both teams agreed that improving the help section could lead to more efficient problem resolution and increased user satisfaction

Data findings

- **42%** of tickets were submitted through the app, while the rest were handled offline
- **22.5%** of users abandoned the ticket-raising process before submitting

Field Visit

We went on a field visit to interview Fasal farmers in different geographical locations to chat about their problems, feelings, and what keeps them going.

User Statements with respect to current help section

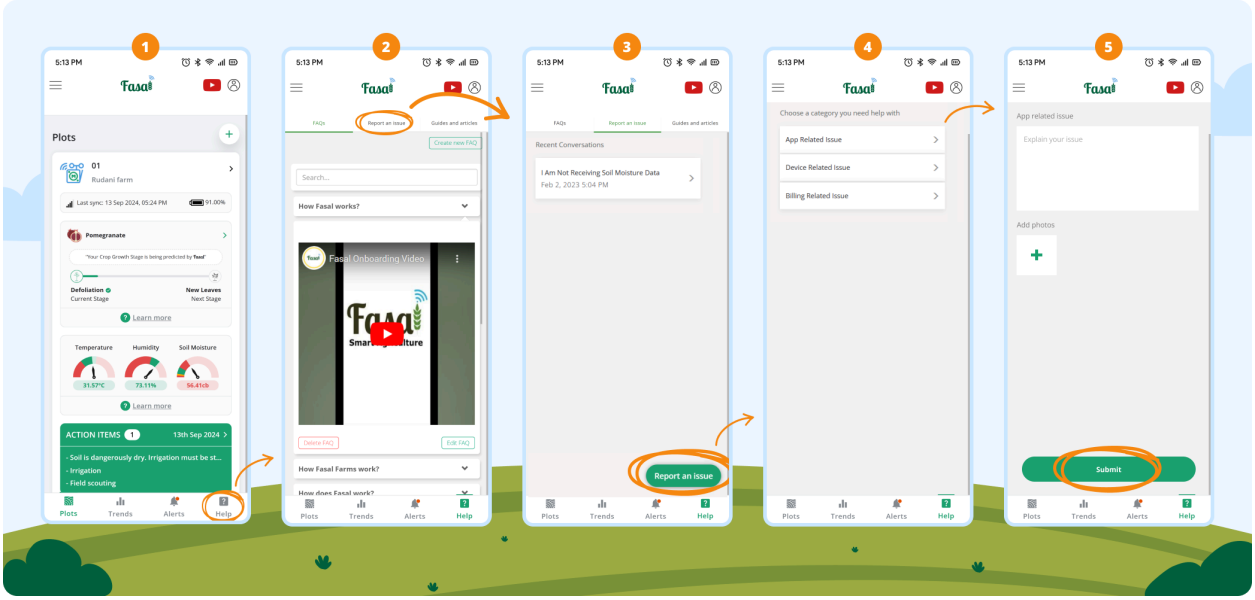
1. Mujhe pata hi nahi hai ki kahan se complaint karte hain, kabhi check nahi kiya.
    a. Translation: I have no idea how to raise a complaint in the app. I've never checked.
2. Mujhe keyboard se type karne mein problem hai. WhatsApp mein bhi sirf audio message bhejta hoon, text message sirf padhta hoon.
    a. Translation: I have difficulty typing on the keyboard. Even on WhatsApp, I only send audio messages and just read text messages.
3. Mujhe di hui categories mein mera issue nahi milta jo main face kar raha hoon.
    a. Translation: I can't find my issue in the given categories in the help section.
4. Sales wale bande ko call karna mere liye easy hai.
    a. Translation: It's easier for me to call the sales person. I can just pick up my phone and call them.

Defining Problem Statement

Problem Statement: Farmers using the Fasal App struggle to effectively report issues due to a complex, text-based help system that doesn't account for language barriers, literacy levels, and user preferences, leading to frustration and inefficient problem resolution.
How Might We Statement: How might we redesign the Fasal App's help system to be more inclusive, user-friendly, and efficient for farmers with varying literacy levels and language preferences, while streamlining the issue reporting and resolution process?

Current Flow & Issues

Currently, farmers with medium to low tech literacy face challenges in discovering how to raise an issue in the Fasal app.



1. **Too many steps:** The current process has 5 steps, which is too long. Many farmers give up before finishing, leaving their problems unsolved. This makes them unhappy with both the Fasal app and device.
2. **Hard to find help:** The button to report a problem is hidden in the help section and isn't easy to see.
3. **Not enough problem types:** The list of problem types doesn't match what farmers actually face. There are too few options, and they don't fit real-world issues well.
4. **Writing is difficult:** Farmers who aren't comfortable with technology find it hard or impossible to explain their problems in writing.
5. **No multilingual support:** The current text-based system doesn't support multiple languages, making it very difficult for farmers to explain issues in their native language using a keyboard. This language barrier can lead to miscommunication and frustration.
6. **More work for support team:** The customer support team has to work harder to understand problems because written explanations are often not clear enough. This makes solving issues take longer and leaves customers less satisfied.

Ideation, Goals & Constraints

We found a problem that affects both our company and the farmers we serve. We visited farms many times to make sure this was a real issue. Then, we started looking for solutions. This shows we care about solving real problems in ways that work well for our users.

1. The solution should be easy to use and interactive, especially for farmers who may not read or write well.
2. Farmers should be able to find it easily in the app.
3. It shouldn't create more work for our customer support team.
4. It should look and feel similar to the rest of the app.

Constraints

1. The ideal solution would be to provide an upfront calling option; however, operational limitations prevented us from implementing this.

> **Introduce audio ticketing feature**
> It will enable the farmers to record their messages/complaint/query via audio.

Design Process

Field Visit & Define

- Interviewed farmers & internal stakeholders to understood their pain points with respect to current help section.

Ideate & Design

- Brainstormed different solutions & presented most viable solution to relevant stakeholders such as Head of product, Product lead etc.

Prototyping & User Testing

- Worked on high fidelity interactive prototype
- Designed a case study for user testing in Useberry.

Feedback Implementation

- Interviewed farmers while they were interacting with prototype.
- Conduct analysis of the user testing report generated by Useberry.

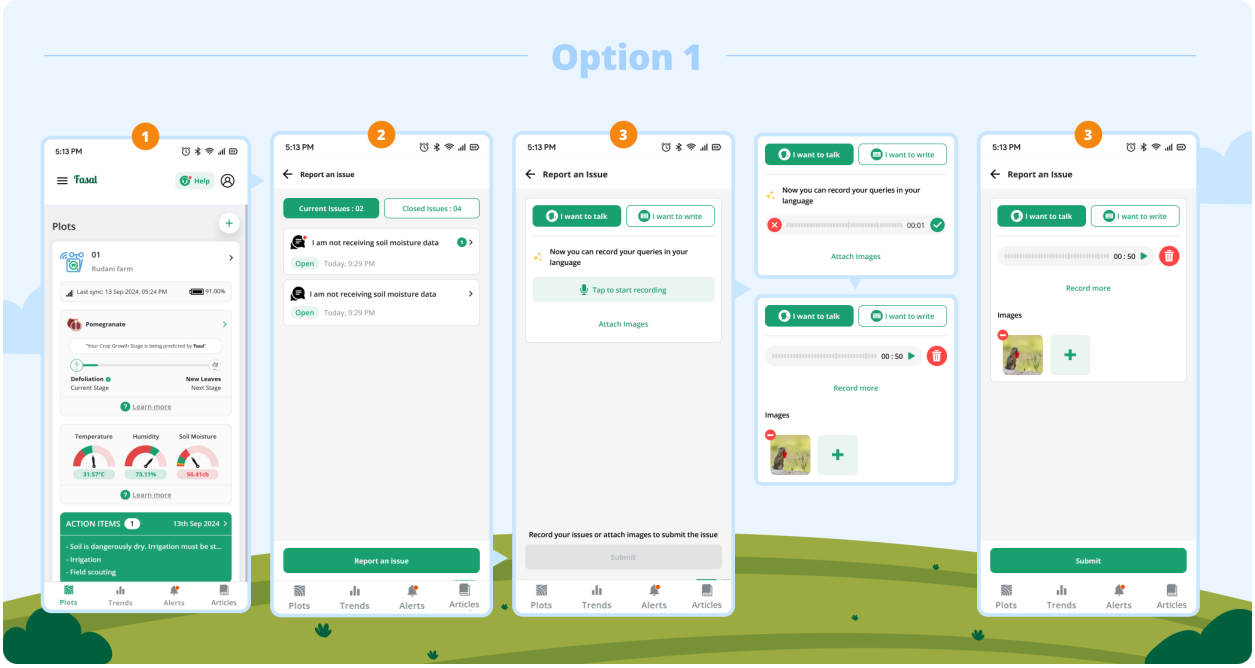
Design Explorations & Finalization

Our goal was to create an easier, more user-friendly interface for reporting issues and getting help. We wanted to simplify the process, make it easier to find things, and add features to help with language and literacy challenges.

> We designed two options for user testing purposes.

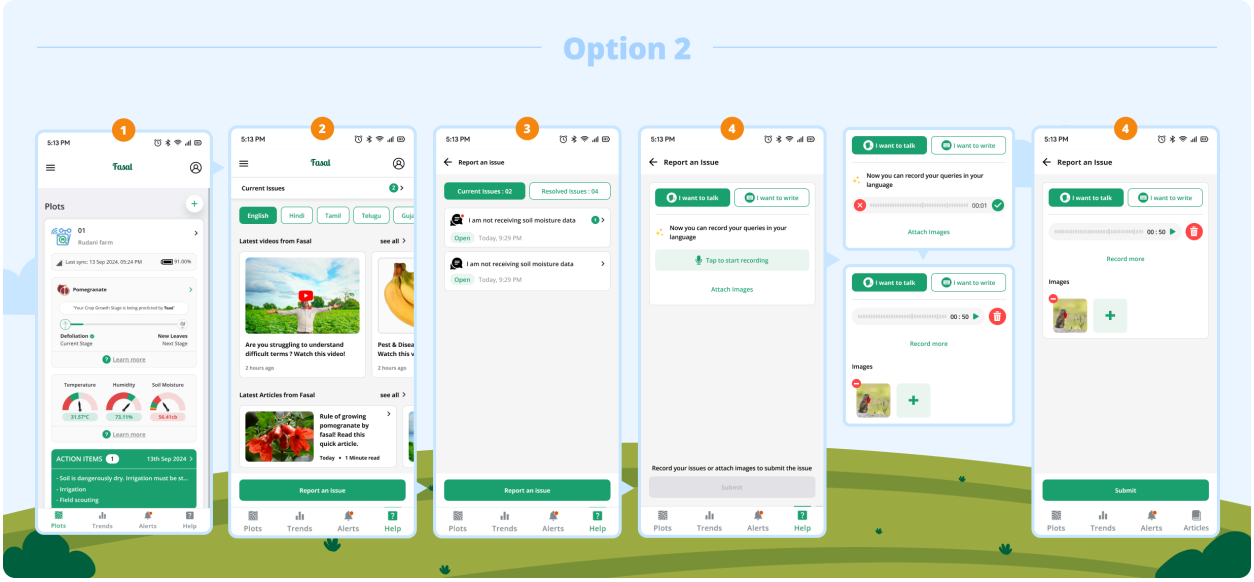
Final design option 1

1. We decided to make the Help CTA separate from articles, more prominent, and easily accessible, instead of hiding it under options.
2. We reduced the number of steps from 5 to 3.
3. We introduced a voice recording option alongside text input, eliminating language barriers.
4. Eliminated predefined problem categories as they didn't accurately reflect real-world issues farmers face.
5. We separated the Articles section from the Help section for improved organization.
6. We improved the overall user interface and experience.



Final design option 2

1. In this option, we kept the Help CTA on the same page as the articles but made it more prominent.
2. We reduced the number of steps from 5 to 3.
3. We introduced a voice recording option alongside text input, eliminating language barriers.
4. Eliminated predefined problem categories as they didn't accurately reflect real-world issues farmers face.
5. We improved the overall user interface and experience.



Prototyping & User Testing

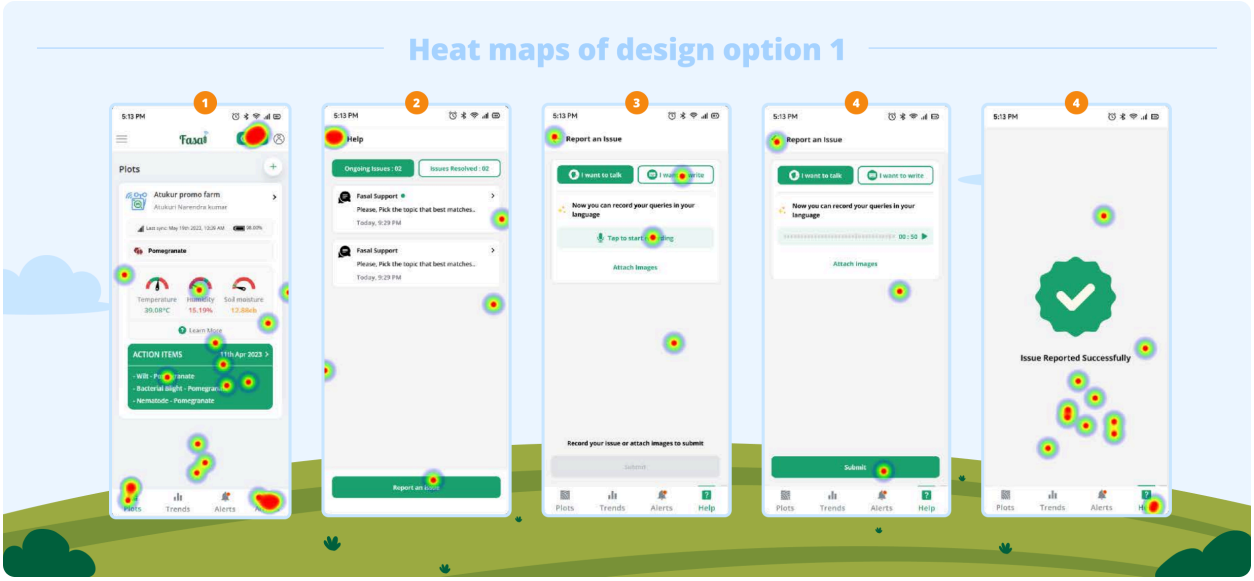
After finalizing two design options, we created interactive prototypes and designed a user test case study in Useberry for further analysis.

We conducted in-person tests of two help section design options with 12 farmers.

Goals of user testing

1. Evaluate the usability and intuitiveness of the new help section design
2. Assess user preference between the two design options
3. Measure the effectiveness of the audio recording feature for issue reporting
4. Gather feedback on the overall user experience and identify any potential areas for improvement

5. Determine if the new design effectively addresses the issues identified in the current flow



Key findings

1. All 12 users successfully completed the flow and were able to raise an issue.
2. Users described the process as very simple and easy to use.
3. Users felt that the call-to-action (CTA) was prominently displayed and easily visible.
4. Four farmers with low tech literacy initially struggled to find resolved issues, but eventually located them.
5. Three farmers found terminology like "Issues resolved" challenging to understand.

Key findings

1. All 12 users successfully completed the flow and were able to raise an issue.
2. Users find it a bit distracting because of the presence of articles and guides in the same section.
3. Eight farmers struggled to find ongoing issues and felt confused due to articles appearing on the same page
4. Three farmers found terminology like "Issues resolved" challenging to understand.

Conclusion

1. **Preference for Design Option 1:** 100% of users found the separate Help CTA in Option 1 more intuitive and easier to access.
2. **Positive response to audio recording:** Users appreciated the ability to record their issues verbally, especially those with limited literacy or typing skills.
    a. Users liked the audio recording feature because they could speak in their own languages.

3. **Reduced steps were well-received:** The streamlined process with fewer steps was perceived as more user-friendly and efficient.
4. **Improved discoverability:** Users reported finding it easier to locate the help section in both new designs compared to the current flow.
5. **Elimination of predefined categories was beneficial:** Users found it easier to report their specific issues without being constrained by preset categories.

Areas for improvement

1. Some terminology needs simplification.
2. Users had difficulty finding resolved issues.

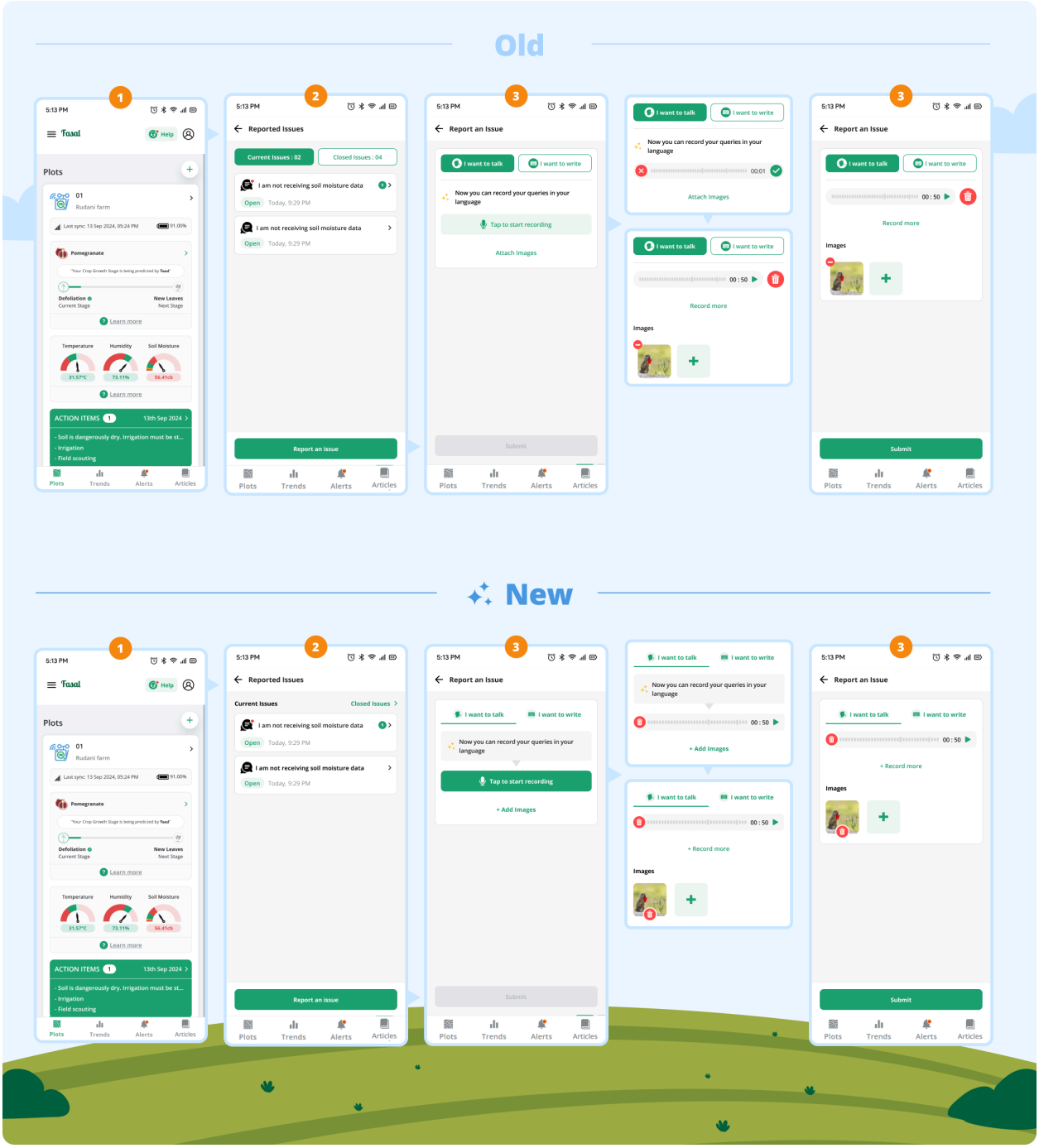
These findings suggest that the new design, particularly Option 1, effectively addresses the main issues identified in the current flow and provides a more accessible and user-friendly experience for farmers using the Fasal app.

Feedback implementation

Based on these findings, we implemented several key changes to improve the user experience:

1. Simplified terminology throughout the help section, using more familiar language for farmers
    a. Changed "Resolved issues" to "Closed issues" for clarity
    b. Changed “Attach images” to “Add images”
2. Redesigned the "Resolved Issues" section to make it more prominent and easier to locate.
    a. We combined resolved and current issues on a single page for easier access.

3. Made the "Tap to record" call-to-action more prominent, as it is the primary action on the page compared to the options "I want to talk" and "I want to write".



These improvements resulted in a more intuitive and user-friendly help section that better serves our farmers' needs. The new design significantly reduced barriers to accessing support, especially for users with limited literacy or technological familiarity. By incorporating audio recording and simplifying the overall process, we've created a more inclusive and efficient system for issue reporting and resolution.

For complete design with all use cases, Please refer to Figma File. (on request basis)

Impact

The implementation of the new Help Section design has led to significant improvements in user experience and operational efficiency:

- **24.9%** increase in tickets submitted through mobile devices
- **107.6%** increase in tickets raised, showing improved user engagement due to better mobile accessibility and the new audio ticketing feature
- **Over a third of users (34.28%)** now use the audio ticketing feature, confirming that language barriers was a real issue.

Way Forward

As we move forward with the improved Help Section for the Fasal App, there are several key areas to focus on:

1. **Continuous User Feedback:** Regularly collect and analyze user feedback to ensure the new design continues to meet farmers' needs and expectations.
2. **Iterative Improvements:** Based on ongoing feedback and performance data, make iterative improvements to further enhance the user experience.

3. **AI Integration:** Explore the possibility of integrating AI-powered voice recognition to automatically categorize and prioritize issues based on audio content.
4. **Knowledge Base Enhancement:** Continuously update and expand the articles and guides section to provide more comprehensive self-help resources for common issues.

By focusing on these areas, we can ensure that the Help Section continues to evolve and provide the best possible support for our farmers, ultimately contributing to the overall success and user satisfaction of the Fasal App.

Thank you

Other case studies

IoT Data Visualization: Simplifying Complex Metrics for 50k+ Users

BazaarNXT: B2B Procurement Platform

Fasal User Research: Decoding Mental Models for Non-Tech Users